Code No: 155BT **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, September - 2021
IMAGE PROCESSING
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) What is digital image processing? Write about the origins of digital image processing.
b) With an example, discuss about gray level digital image processing. [7+8]

2.a) Discuss about the basic concepts in sampling and quantization.
b) What is the principle objective of sharpening spatial filters? [7+8]

3. Illustrate on histogram processing for image enhancement with example. [15]

4.a) Describe about image restoration degradation model.
b) Briefly discuss about inverse filtering. [8+7]

5.a) What is the concept behind algebraic approach to restoration- Explain?
b) With an example, give the difference between enhancement and restoration. [8+7]

6. Summarize the detection of edge, line, and point based on segmentation methods with suitable examples. [15]

7.a) Demonstrate the fundamentals for image compression.
b) Discuss about any two-coding methods, used for basic compression. [7+8]

8. Write short notes on the following.
a) KLT and SVD 2D Transformations.
b) Lossy compression. [8+7]

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